

Consider the TM $M = (Q, \Sigma, s, \delta)$ where

$Q = \{s, q, q'\}$, $\Sigma = \{_, 0, 1\}$ and the transition function

$\delta \in (Q \times \Sigma) \rightarrow (Q \times \Sigma \times \{L, R\})$ is given by:

δ	$_$	0	1
s	$(q, _, R)$		
q	$(q', 0, L)$	$(q, 1, R)$	$(q, 1, R)$
q'			$(q', 1, L)$